

Implementing Verdier's robust extrapolation in the CKT model

Relaxing the i.i.d. assumption on ν_{it}^l

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Our current estimator corresponds to Verdier's simple extrapolation

We currently follow Suri (2011) and Tjernström et al. (2026):

- impose the LCA restriction $\Delta_i = \beta + \phi \theta_i$
- identify ϕ from switcher trajectories
- extrapolate to non-migrants (d_N, d_T) along the LCA line

Verdier (2020) treats this class of methods as a “simple extrapolation” & proposes a robust alternative. Our estimator and his simple version are not literally the same procedure (different aggregation level, reciprocal slope convention) but target the same LCA relationship asymptotically.

The simple extrapolation assumes i.i.d. non-pecuniary shocks

Our current Assumption A2 states that the non-pecuniary utility shock

ν_{it}^l is i.i.d. across individuals i , time t , & locations l

This implies:

- ν_{it}^l independent of baseline skill & comparative advantage **everywhere**
- rules out systematic differences in ν_{it}^l across regions/institutions
- workers in different locations have the same shock distribution, conditional on comparative advantage

The migration literature documents what A2 rules out

Regional heterogeneity is a central fact of this literature:

- Bryan and Morten (2019): structural model of Indonesian internal migration estimates large bilateral migration costs
- Lagakos et al. (2020): migration cost estimates differ across developing countries
- Pulido and Świącki (2021): cross-sector barriers in Indonesia explain a large share of within-country income gaps
- China: hukou is an institutional source of region-specific migration barriers

Our current model rules this out by assumption

Verdier's robust extrapolation weakens A2 to a conditional version

Decompose the non-pecuniary shock:

$$\nu_{it}^l = m_l(v_i) + \tilde{\nu}_{it}^l$$

where v_i is an observed, time-invariant individual-level index

New assumption (A2'):

- $\tilde{\nu}_{it}^l$ is i.i.d. *conditional on* v_i
- $m_l(v_i)$ can correlate arbitrarily with comparative advantage

We use first-wave province as v_i to index region-level shocks (all three countries)

Why this is weaker

A2 (our assumption) requires *unconditional* independence

A2' (VV's assumption) requires only *within-province* independence

In practice: the LCA gets a cluster-specific intercept

Slope stays common but the intercept varies

Simple:

$$\Delta_i = \beta + \phi \theta_i$$

Robust:

$$\Delta_i = \beta(v_i) + \phi \theta_i$$

- slope ϕ remains common; this is still the LCA restriction
- intercept $\beta(v_i)$ absorbs cluster-level $m_l(v_i)$ heterogeneity
- ϕ identifies off within-cluster switcher variation
- extrapolation to non-migrants happens cluster-by-cluster, then averaged

v_i is generic in VV's framework (village in his Suri-Kenya application; for CKT, see slide on geography below).

Pooling provinces mixes differentially-selected switcher samples

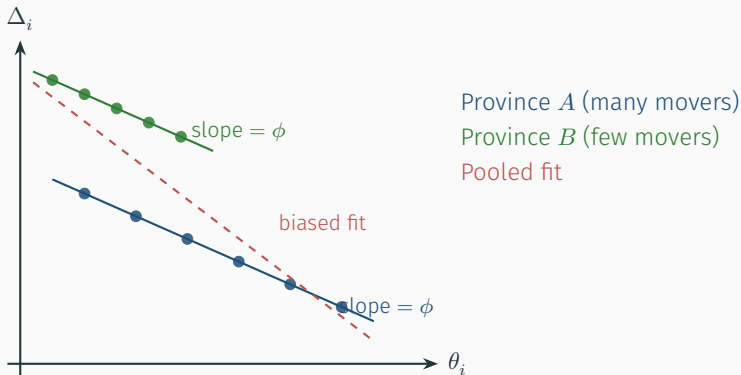
Consider two provinces:

- **Province A** (many movers): wide switcher sample, broad range of θ_i
- **Province B** (few movers): skewed sample, only high-return types migrate

Pooled regression: can be biased by across-province difference in selection

Within-province: clean slope, holds region-level shocks constant

Within-province slopes agree—the pooled slope is pulled off



Each province's switcher cloud has true slope ϕ

The pooled line tilts because of intercept differences, not because ϕ varies

How Suri, Verdier, and CKT
identify the LCA slope

Suri (2011) identifies ϕ via panel SUR & minimum distance

Suri (2011)'s two-stage indirect approach on a multi-year, binary-adoption panel:

Stage 1 (reduced form). Stack one outcome equation per year. SUR over the panel:

$$y_{it} = \pi_{0,t} + \sum_{\underline{d} \in \mathcal{D}} \pi_{\underline{d},t} \mathbb{1}\{\underline{d}_i = \underline{d}\} + x'_{it} \gamma_t + u_{it}$$

Each coefficient $\pi_{\underline{d},t}$ is the average y in adoption pattern $\underline{d}$ in year t

Stage 2 (structural / OMD). Express each $\pi_{\underline{d},t}$ as a known function of $(\beta, \phi, \lambda_1, \dots, \lambda_K)$ implied by the CRC restriction

$$g_{\underline{d},t}(\beta, \phi, \lambda) = \pi_{\underline{d},t}$$

Solve via weighted minimum distance: $\hat{\theta} = \arg \min (\hat{\pi} - g(\theta))' W (\hat{\pi} - g(\theta))$

- Identification: SUR over-identifies the structural parameters; ϕ gets pinned by the across-equation cross-restrictions
- Implemented in our `randcoef.ado`, which replicates Suri's estimation approach

VV Step 1: worker FE plus worker-by-treatment interactions

Verdier (2020)'s CRC model (his eq. 2.4):

$$y_{it} = a_i + b_i D_{it} + f_t + x'_{it} \gamma + u_{it}$$

Worker-level intercept a_i (rural mean); worker-level treatment effect b_i (return).

Step 1 estimates this by OLS on n worker indicators (absorbing a_i) plus n worker-by-treatment interactions (recovering b_i), with period indicators / covariates.

For each mover, this delivers $\hat{a}_i$ and $\hat{a}_i + \hat{b}_i$ as the within- i regime means of $(y - x' \hat{\gamma})$:

$$\hat{a}_i = \text{avg of } (y - x' \hat{\gamma}) \text{ over } D_{it} = 0 \text{ periods,}$$

$$\hat{a}_i + \hat{b}_i = \text{avg of } (y - x' \hat{\gamma}) \text{ over } D_{it} = 1 \text{ periods.}$$

Then $\hat{b}_i = (\hat{a}_i + \hat{b}_i) - \hat{a}_i$.

Movers have both; untreated stayers only $\hat{a}_i$; treated stayers only $\hat{a}_i + \hat{b}_i$.

VV Step 2: IV regression of $\hat{a}_i$ on $\hat{b}_i$ identifies α_1

VV writes the LCA in the direction *a-on-b*:

$$a_i = \alpha_0 + \alpha_1 b_i + e_i$$

Naive OLS is biased: $\hat{b}_i$ is noisy, so it is a regressor with measurement error.

VV instruments $\hat{b}_i$ with the binary trajectory pattern $(x_{i1}, \dots, x_{iT})$:

$$E[x_{it} (\hat{a}_i - \alpha_0 - \alpha_1 \hat{b}_i)] = 0, \quad t = 1, \dots, T$$

Robust version (cluster-level cost shifters): add cluster- v_i FE to the IV regression. By Frisch-Waugh, demean LHS, RHS, *and* instrument within v_i ; intercept drops out:

$$E\left[(x_{it} - \bar{x}_{v_i,t}) ((\hat{a}_i - \bar{\hat{a}}_{v_i}) - \alpha_1 (\hat{b}_i - \bar{\hat{b}}_{v_i}))\right] = 0$$

Run on movers only. Cluster-robust SE at v_i .

CKT identifies ϕ at the trajectory level

Trajectory pooling: every worker with $\underline{d}_i = \underline{d}$ has the same expected baseline

$$E[\theta_i \mid \underline{d}_i = \underline{d}] = \mu_{\underline{d}} - \mu_{\underline{d}_0}$$

Single GMM equation, one scalar $\mu_{\underline{d}}$ per trajectory:

$$y_{it} = \sum_{\underline{d}} \mu_{\underline{d}} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \Delta_{\underline{d}_0} D_{it} + \phi \sum_{\underline{d} \in \mathcal{D}_S \setminus \underline{d}_0} (\mu_{\underline{d}} - \mu_{\underline{d}_0}) D_{it} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \dots$$

Identification: switcher $\times$ choice indicators as instruments

Robust variant: same equation, switcher $\times$ choice instruments *village-demeaned* (like VV)

Both procedures collapse to the same trajectory-level moments

VV's IV groups movers by trajectory pattern g and uses within-group means $(\bar{\hat{a}}_g, \bar{\hat{b}}_g)$. Within-group averaging washes out the Step 1 measurement error in $\hat{b}_i$. With $T = 2$ and two mover bins $(0, 1), (1, 0)$, the IV reduces to a Wald estimator:

$$\hat{\alpha}_1 = \frac{\bar{\hat{a}}_{(0,1)} - \bar{\hat{a}}_{(1,0)}}{\bar{\hat{b}}_{(0,1)} - \bar{\hat{b}}_{(1,0)}}$$

CKT computes per-trajectory $\mu_{\underline{d}}$ and $\Delta_{\underline{d}}$ directly via GMM. Same set of trajectory-level points; both procedures fit a line through them.

- Without cluster structure: VV-simple and CKT-simple are equivalent up to slope-direction convention.
- With cluster structure: CKT-robust pools *within each cluster* (assumption A3 below). VV-robust uses only within-cluster variation, no across-cluster pooling needed.

VV's α_1 and CKT's ϕ target different population scalars

VV's LCA: $a_i = \alpha_0 + \alpha_1 b_i + e_i$ (α_1 : slope of a on b)

CKT's LCA: $\Delta_i = \beta + \phi \theta_i + \xi_i$ (ϕ : slope of Δ on θ)

Two distinctions:

- **Regression direction.** Slope of a on b vs slope of Δ on θ . Reciprocal only when $|\text{Corr}(a_i, b_i)| = 1$ (perfect linear LCA). Sign agrees automatically when both target the same correlation; magnitudes differ.
- **Right-hand-side variable.** Rural baseline a_i vs comparative advantage $\theta_i = b_R(\theta_i^U - \theta_i^R)$. CKT's structural model interprets Δ_i in terms of two skills (θ_i^U, θ_i^R) ; VV is agnostic about that structure. Both target the same scalar empirically (under correct assumptions); CKT's two-skill structure buys interpretation, not identification.

Special-case identity: when $\xi_i = e_i = 0$ and $\theta_i \equiv a_i$ (deterministic, single-skill limit), $\alpha_1 = 1/\phi$. Outside that limit, $\alpha_1 \neq \phi$ and $\alpha_1 \neq 1/\phi$. See appendix for the algebra.

Three estimators, three assumption regimes

Assumptions:

A1 LCA at the worker level: $\Delta_i = \beta(v_i) + \phi \theta_i + \xi_i$.

A2 Unconditional: ν_{it}^l i.i.d. across workers (no cluster heterogeneity).

A2' Within-cluster only (ν_{it}^l may correlate with v_i); strictly weaker than A2.

A3 Trajectory pooling: $\theta_i \mid \underline{d}_i$ same distribution in every cluster.

Estimator	Needs	Targets
CKT main (<code>run_grc</code> , Suri-style)	A1, A2	ϕ
CKT robust-vv (<code>run_grc_robust_vv</code>)	A1, A2', A3	ϕ
VV worker-level (robust)	A1, A2'	α_1

Under A2 (stronger): all three target the same LCA relationship. Under A2' only: CKT main is biased. When A3 fails: CKT robust-vv picks up a bias of $\text{Cov}_s(\bar{\beta}(s), \mu_{\underline{d}} - \mu_{\underline{d}_0})$; VV stays consistent. $\hat{\alpha}_1^{\text{VV}}$ and $\hat{\phi}$ differ by convention (previous slide); reciprocal under deterministic LCA.

Our implementation moves cluster structure to the instruments

`run_grc_robust_vv`: keep CKT's parameters, swap the instruments

- Regress $\mathbb{1}\{\underline{d}_i = s\} D_{it}$ on $\mathbb{1}\{v_i\}$ among trajectory- s workers
- Use residuals as instruments for ϕ identification

Same parameter set as the simple GRC ($\mu_{\underline{d}}, \Delta_{\underline{d}_0}, \kappa, \gamma$); no $\beta(v)$ parameters

The move buys a weaker assumption at tractable cost

Pros:

- $A2'$ matches what the migration literature documents
- the CHN hukou split acts as a principled special case ($v_i = \text{hukou}$) rather than an ad hoc fix (unless VV “fixes” without needing to bring in hukou)
- stayer ATE estimates defensible against Verdier’s concern in Suri’s data

Cons:

- Δ_{d_N} averages over provinces with switcher support, not the full population
- thinner identification of ϕ : between-province switcher variation no longer contributes; only the within-province component does
- cluster-robust standard errors at the v_i level

Where the worker-level estimator
can fail

VV uses two geographic levels; we currently use only one

VV (Suri-Kenya). Two geographic units, both used:

- Province: enters as province $\times$ year FE in Step 1 covariates (Verdier, 2020, §4.1).
- Village: cluster index v_i for robust extrapolation (§4.3). Suri's data: ~ 94 villages, ~ 12 farmers each.

CKT processed data. Three sub-province levels exist in our panels:

- CHN: `cid` (community), $\sim 4,066$ distinct, median ~ 2 workers.
- IDN: `keca` (kecamatan / subdistrict), $\sim 1,669$ distinct, median ~ 4 .
- TZA: `ward`, ~ 135 distinct, median ~ 97 .

What we currently do, and why it's wrong

Our vanilla VV port clusters at first-wave province (`vfirst`, ~ 25 provinces). That choice came from a misreading: we saw “province” in VV’s Step 1 and missed that his v_i in Step 2 is the village. The fix is to rerun at the village-equivalent level (`cid` / `keca` / `ward`), keeping province as a Step 1 covariate per VV §4.1.

Cluster-FE IV is fragile when within-cluster movers are scarce

Identification needs (i) within-cluster movers, (ii) trajectory variation among them, (iii) cluster size large enough for FE-IV to be well-conditioned.

Fragility cases:

- zero movers in a cluster $\Rightarrow$ cluster-residualized period-treatment indicators are zero, no contribution to identification;
- one or two movers $\Rightarrow$ within-cluster demeaning eats most of the variation;
- with T periods and k movers, the within-cluster trajectory design has rank at most $\min(T, k - 1)$ after partialling worker FEs. Clusters with $k \leq T$ contribute fewer free moments than nominal.

This is a general property of cluster-FE IV with worker-level fixed effects. The binding question is not “is V too large?” but “how many clusters have too few movers?”

An observed shifter w_i provides a separate falsification test (pending)

Regress estimated θ_i^* on Δ_i and an observed shifter w_i :

$$\theta_i^* = \alpha_0 + \alpha_1 \Delta_i + \alpha_2 w_i + e_i, \quad H_0 : \alpha_2 = 0$$

Kenya: w_i = distance to nearest seed seller; rejects if $\alpha_2 \neq 0$

w_i must be observed, exogenous to skill conditional on Δ_i , and correlated with ν_{it}^l

Candidates in our data:

- distance / travel time to nearest urban center
- rainfall variability or pre-period climate shock (mergeable via GPS/admin code)
- distance to family already in urban area (closest analogue, but reflection problem)

None is perfect; data availability is the binding constraint

Appendix: parameter mapping

Object-by-object map: parameters and LCA structure

Object	VV	CKT
<i>Worker-level objects</i> (parameterized only in VV)		
Worker rural mean	$\hat{a}_i$	not parameterized
Worker urban mean	$\hat{a}_i + \hat{b}_i$	not parameterized
Worker return	$\hat{b}_i$	not parameterized
<i>Trajectory-level objects</i> (CKT's natural level)		
Trajectory rural mean	$\bar{\bar{a}}_g$ (post-Step-1 avg in bin g)	$\mu_{\underline{d}}$ (joint GMM parameter)
Trajectory return	$\bar{\bar{b}}_g$	$\Delta_{\underline{d}}$ (joint GMM parameter)
<i>LCA structure</i>		
LCA equation	$a_i = \alpha_0 + \alpha_1 b_i + e_i$	$\Delta_i = \beta + \phi \theta_i + \xi_i$
Slope parameter	α_1 (slope of a on b)	ϕ (slope of Δ on θ)
Right-hand-side variable	rural baseline a_i	comparative advantage θ_i

Object-by-object map: geographic structure and estimator

Object	VV	CKT-main (<code>run_grc</code>)	CKT-vv (<code>run_grc_robust_vv</code>)
<i>Geographic / clustering structure</i>			
Province (covariate use)	province×year FE in Step 1	not used as a separate level	not used as a separate level
Cluster index v_i	village (Suri-Kenya, ~ 94, ~ 12 farmers each)	none (no clustering)	first-wave province as currently configured (~ 25); to be switched to community / kecamatan / ward
<i>Estimator</i>			
Step 1 estimator	OLS w/ worker×regime FE	n/a (joint GMM)	n/a (joint GMM)
Step 2 estimator	IV: $\hat{a}$ on $\hat{b}$, cluster FE	joint GMM with switcher×choice instru- ments	joint GMM with <i>cluster-residualized</i> switcher×choice instruments
Aggregation level	worker (movers only)	trajectory	trajectory
Inference	cluster bootstrap or analytical, clustered at v_i	default GMM SE	cluster-robust at v_i

When the slopes coincide, and when they don't

Plain English. Suppose every worker's return fell *exactly* on the LCA line (no noise) and there were one underlying skill so that rural baseline a_i and comparative advantage θ_i were the same object up to a constant. Then Δ_i would be a perfect linear function of a_i and the inverse would also be linear with reciprocal slope.

Algebra. $\Delta_i = \beta + \phi a_i$ inverts to $a_i = -\beta/\phi + (1/\phi)\Delta_i$, so $\alpha_1 = 1/\phi$.

Two ways the identity breaks.

- *LCA noise.* OLS slope of a on b is $\text{Corr}(a, b) \sigma(a)/\sigma(b)$; slope of b on a is $\text{Corr}(a, b) \sigma(b)/\sigma(a)$. Product equals $\text{Corr}(a_i, b_i)^2 \leq 1$. Reciprocal only when $|\text{Corr}(a_i, b_i)| = 1$.
- *Two-skill structure.* $\theta_i = b_R(\theta_i^U - \theta_i^R)$ uses both skills; a_i uses θ_i^R alone. Slope of Δ on $\theta \neq$ slope of Δ on a .

Outside the special-case limit, $\alpha_1 \neq \phi$ and $\alpha_1 \neq 1/\phi$.

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